Linear Programming Approach to the Graph Square Problem

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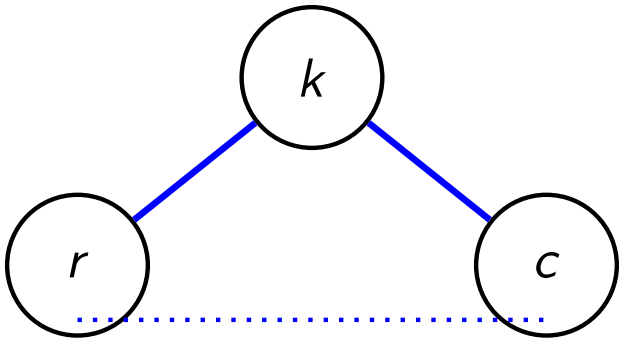
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Square of a Graph

Two-step connections

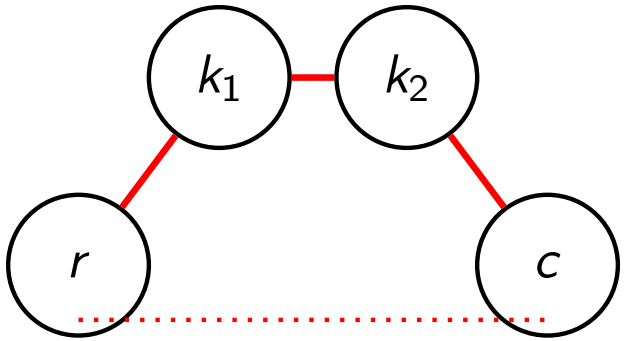


two steps

$$(A^2)_{rc} = \sum_{k=1}^{n} A_{rk} A_{kc}.$$

Counts walks of length two from r to c.

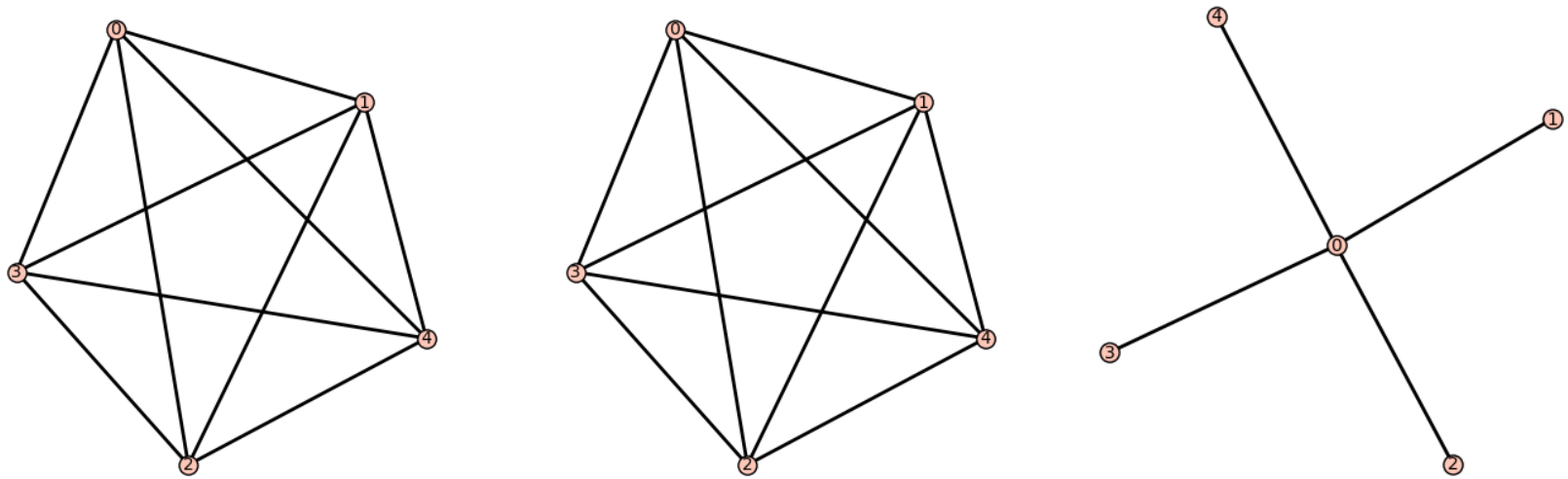
Three-step connections



three steps

$$(A^3)_{rc} = \sum_{k=1}^{n} A_{rk} (A^2)_{kc}.$$

Counts walks of length three from r to c.



(a) K_5. (b) K_5, a root of K_5. (c) $K_{1,4}$, a root of K_5.

Figure: A few of the 15 non-isomorphic graph roots of K_5.

The Graph Square Problem

Problem (SQUARE)

Instance: *A finite, simple, undirected graph* $G = (V, E)$.

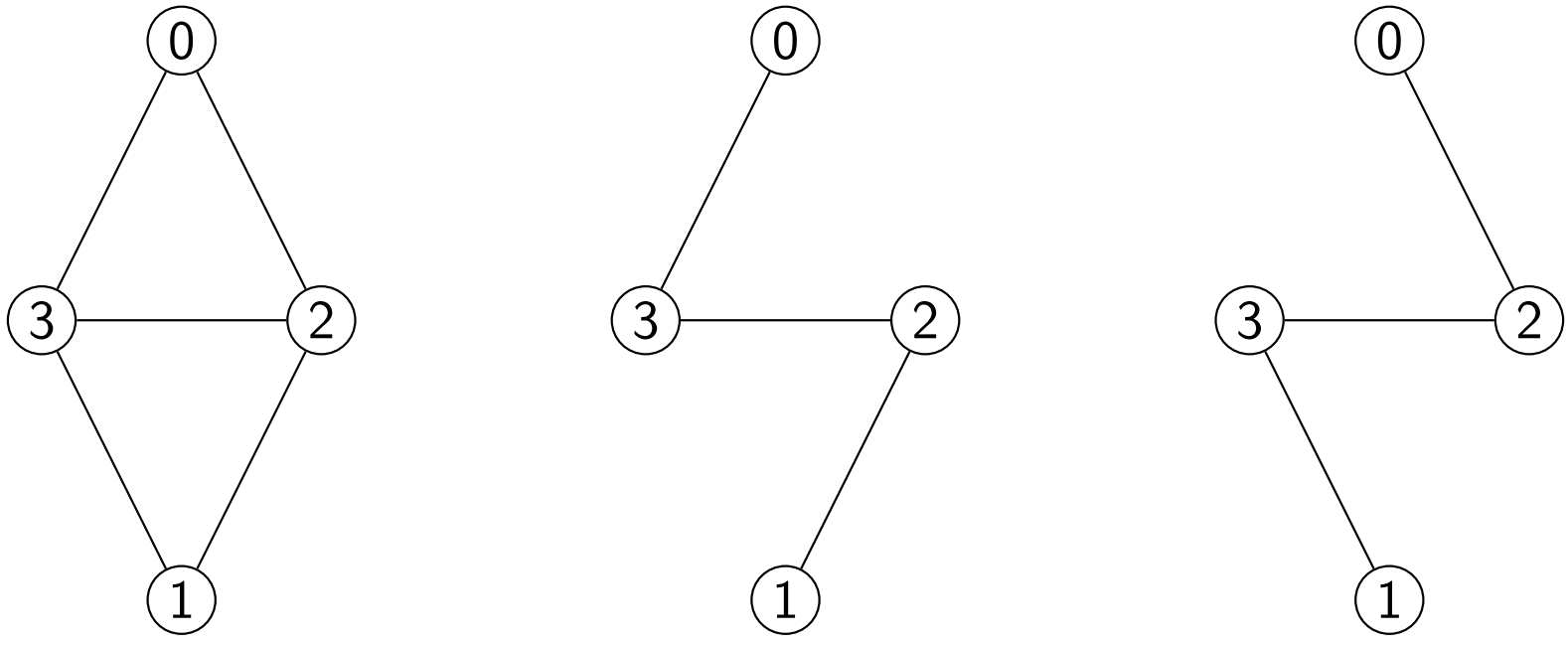
Question: *Does there exist a finite, simple, undirected graph* H *such that*

$$G = H^2?$$

Let the **graph root** $H = (V, E_H)$ be a finite, simple, undirected graph, where V is the vertex set with $|V| = n$, and $E_H \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$ is the edge set. The **square of a graph** H, denoted by $H^2 = (V, E_{H^2})$ is the graph on the same vertex set with edge set

$$E_{H^2} = \{uv \mid u \neq v, 1 \leq d_H(u, v) \leq 2\}. \tag{1}$$

When $G = H^2$ we say that H is a **root** of G.



(a) G (b) T_1 (c) T_2

Figure: The graph G has two possible P_4 square roots that are trees, T_1 and T_2. The two roots are isomorphic, differing only by the interchange of vertices 2 and 3.

NP-Completeness

Motwani and Sudan (1994) proved that the graph square problem is **NP**-complete by showing that the problem is **NP**-hard through a polynomial reduction from Not-All-Equal 3-SAT.

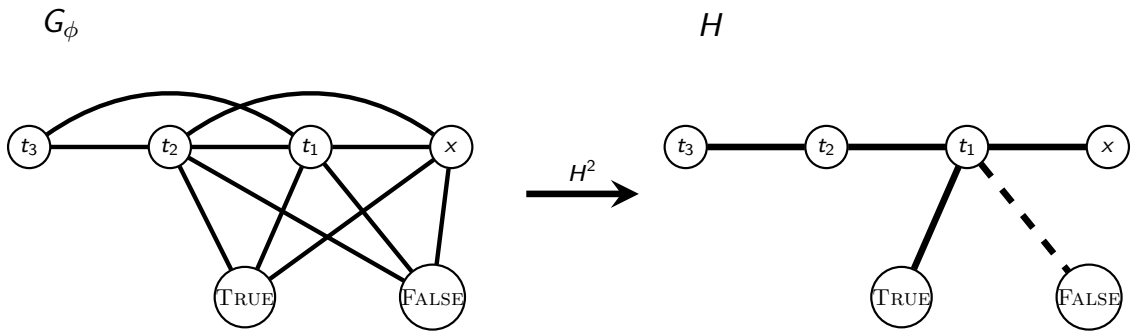
Problem (NAESAT)

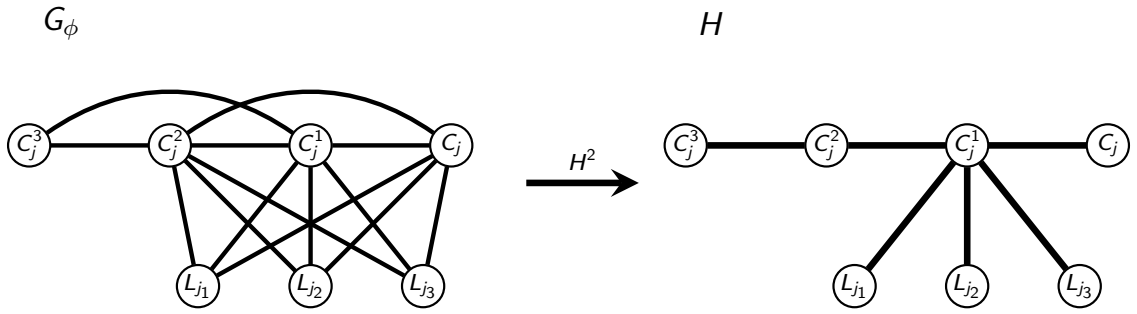
Instance: *A 3-CNF formula*

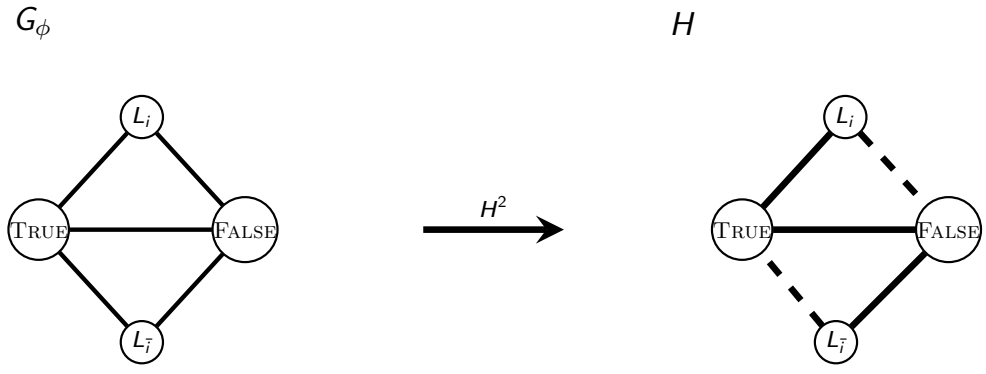
$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$$

where each clause C_i is over variables $x_1, \ldots, x_n$.

Question: *Does there exist an assignment of* TRUE *and* FALSE *to the variables such that every clause contains at least one* TRUE *literal and at least one* FALSE *literal?*


G_ϕ
H
t_3
t_2
t_1
x
True
False
H^2


G_ϕ
H
C_j^3
C_j^2
C_j^1
C_j
L_{j_1}
L_{j_2}
L_{j_3}
H^2


G_ϕ
H
L_i
True
False
$L_{\bar{i}}$
H^2

Restricting the Square vs the Root

Square G	**Root H**	**Main result**
Unrestricted	Trees	Ross and Harary (1960): structural characterization and reconstruction of tree roots, unique up to isomorphism [RH60].
Unrestricted	Trees	Lin and Skiena (1995): optimal linear-time recognition and reconstruction of tree roots [LS95].
2-connected	Unrestricted	Fleischner (1974): the square of a 2-connected graph has a Hamiltonian cycle [Fle74].
Interval squares	Unrestricted	Lau and Corneil (2004): polynomial-time construction of roots from interval squares [LC04].
Unrestricted	Block graphs	Le and Tuy (2007): linear-time recognition of roots of block graphs [LT07].
Unrestricted	Bipartite	Lau (2006): linear-time construction of bipartite roots [Lau06].
Directed graphs	Directed graphs	Geller (1970): necessary and sufficient conditions for directed square-root existence [Gel70].
Graphs of prescribed girth	Unrestricted	Farzad, Lau, Le, and Tuy (2009) and Farzad and Karimi (2012): complexity dichotomy for square-root recognition [Far+09; FK12].
Maximum degree ≤ 6	Unrestricted	Cochefert, Couturier, Golovach, Kratsch, and Paulusma (2013): minimum-edge roots in polynomial time and maximal-edge roots in exponential time [Coc+13].
Unrestricted	Oriented graphs, maximum degree ≤ 3, girth ≥ 4	Duffy (2025): classification of squares of oriented graphs under these root restrictions [Duf25].

Maximal Cliques and Unicliqual Vertices in Tree Roots

Input: A graph G where $G = T^2$
Output: A tree T
Step 1
if $G = K_p$ is complete **then**
    Return tree T with depth 1, root any vertex
else
    Continue
end if
Step 2
Identify all maximal cliques $C_1, \ldots, C_n$ of G (e.g. Bron–Kerbosch [BK73])
Step 3
Let $b_1, \ldots, b_n$ be multicliqual points for $C_1, \ldots, C_n$
Build graph T on $\{b_1, \ldots, b_n\}$
for each pair (C_i, C_j) with $|C_i \cap C_j| = 2$ **do**
    Add edge (b_i, b_j) in T
end for
T is a tree
Step 4
for each clique C_i **do**
    Let n_i be the number of unicliqual points in C_i
    Attach n_i leaves to b_i in T
end for
Resulting T has $|V(G)|$ nodes and $T^2 = G$

Integer Program

Equation (1) is equivalent to two distinct vertices are adjacent in $G = H^2$ if and only if they are connected by a path of length one or two in H.

$$A_G = (A_H)^2 \vee A_H. \tag{2}$$

$$(A_H)^2_{ij} = \big((A_H)_{i1} \wedge (A_H)_{1j}\big) \vee \big((A_H)_{i2} \wedge (A_H)_{2j}\big) \vee \cdots \vee \big((A_H)_{in} \wedge (A_H)_{nj}\big)$$

$$g_{ij} \text{ if and only if } \Big(\bigwedge_k (x_{ik} \wedge x_{kj}) \vee x_{ij}\Big). \tag{3}$$

Lemma

Let G and H be graphs with adjacency matrices A_G and A_H,respectively. The graph H is a root of G if and only if

$$A_G = \mathbb{1}_{\geq 1}\big((A_H)^2 - D_H + A_H\big).$$

i) $x_{ij} \in \{0,1\}$ denotes the (i,j) entry of the adjacency matrix A_H.

ii) $y_{ikj} \in \{0,1\}$ represents the product $x_{ik}x_{kj}$ in $(A_H)^2_{ij}$.

iii) $z_{ij} \in \mathbb{Z}_{\geq 0}$ represents the argument of the indicator function.

iv) $M \in \mathbb{Z}_{>0}$ be a sufficiently large constant $M \leq n(n-1) = O(n^2)$.

$$D_H = \begin{bmatrix} \deg_H(1) & 0 & \cdots & 0 & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ \vdots & \cdots & \deg_H(i) & \cdots & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & \deg_H(n) \end{bmatrix}, \quad A_H = \begin{bmatrix} x_{11} & \cdots & x_{1j} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & \cdots & x_{ij} & \cdots & x_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nj} & \cdots & x_{nn} \end{bmatrix}$$

$$(A_H)^2 = \begin{bmatrix} \sum_k x_{1k}x_{k1} & \cdots & \sum_k x_{1k}x_{kj} & \cdots & \sum_k x_{1k}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_k x_{ik}x_{k1} & \cdots & \sum_k x_{ik}x_{kj} & \cdots & \sum_k x_{ik}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_k x_{nk}x_{k1} & \cdots & \sum_k x_{nk}x_{kj} & \cdots & \sum_k x_{nk}x_{kn} \end{bmatrix}.$$

$$\begin{bmatrix} g_{11} \\ \vdots \\ g_{1j} \\ \vdots \\ g_{1n} \\ \vdots \\ g_{ij} \\ \vdots \\ g_{nn} \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}(z_{11}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1j}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1n}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{ij}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{nn}) \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{1k}x_{k1} - (D_H)_{11} + x_{11}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{1k}x_{kj} - (D_H)_{1j} + x_{1j}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{1k}x_{kn} - (D_H)_{1n} + x_{1n}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{ik}x_{kj} - (D_H)_{ij} + x_{ij}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{nk}x_{kn} - (D_H)_{nn} + x_{nn}\big) \end{bmatrix}. \tag{4}$$

Objective Function:

$$\max \quad 0 \tag{5}$$

Variables:

$$x_{ij} \in \mathbb{Z}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{6}$$

$$y_{ikj} \in \mathbb{Z}_{\geq 0}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{7}$$

$$z_{ij} \in \mathbb{Z}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{8}$$

Constraints:

(1) Indicator argument:

$$z_{ii} = \sum_{k=1}^{n} y_{iki} - \sum_{k=1}^{n} x_{ik}, \forall i \in \{1, \ldots, n\}, \tag{9}$$

$$z_{ij} = \sum_{k=1}^{n} y_{ikj} + x_{ij}, \qquad \forall i \neq j \in \{1, \ldots, n\}, \tag{10}$$

(2) Indicator function:

$$z_{ij} \leq M g_{ij}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{11}$$

$$z_{ij} \geq g_{ij}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{12}$$

(3) Product:

$$y_{ikj} \leq x_{ik}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{13}$$

$$y_{ikj} \leq x_{kj}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{14}$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{15}$$

(4) Upper Bound:

$$x_{ij} \leq 1, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{16}$$

$$y_{ikj} \leq 1, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{17}$$

(5) Symmetry:

$$x_{ij} = x_{ji}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{18}$$

(6) Diagonal:

$$x_{ii} = 0, \qquad \forall i \in \{1, \ldots, n\}. \tag{19}$$

The integer program on the arbitrary graph G using Equations (6)–(19) together with the constant objective function Equation (5) is denoted as $\mathcal{IP}(G)$.
The total number of simple graphs on $n = 2, 3, 4, \ldots$ unlabeled nodes is the sequence $2, 4, 11, 34, 156, 1044, 12346, \ldots$ given by A000088. The number of graphs which are squares is the sequence $2, 3, 6, 11, 28, 77, 307, \ldots$ given by A382181.

n	H exists	H does not exist
3	3	1
4	6	5
5	11	23
6	28	128
7	77	967
8	307	12039

Table: Outcomes of the integer program for graphs of different orders of n.

Linear Relaxation

$$x_{ij} \in \mathbb{R}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{20}$$

$$y_{ikj} \in \mathbb{R}_{\geq 0}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{21}$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}. \tag{22}$$

Such a linear program may have *integral solutions* and *fractional solutions* when the root exists and *fractional solutions* and *infeasible solutions* when the root does not exist.

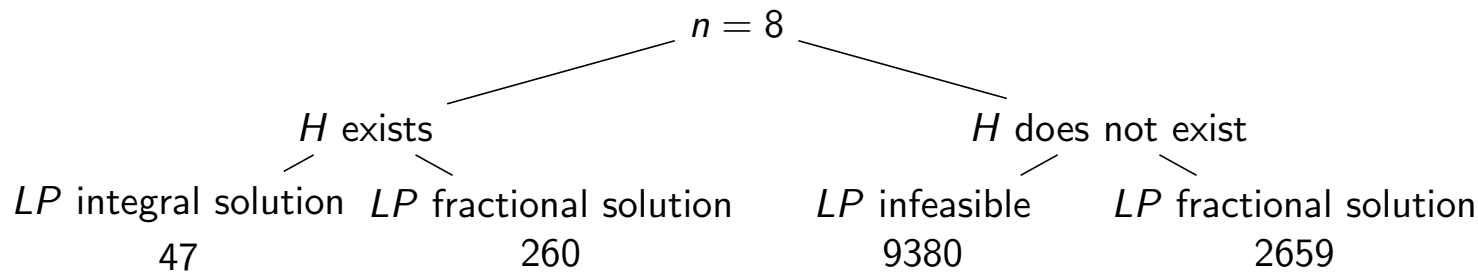
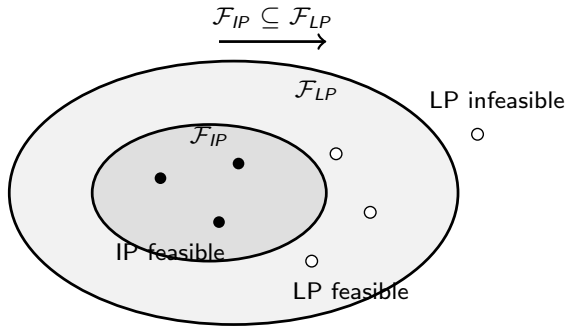
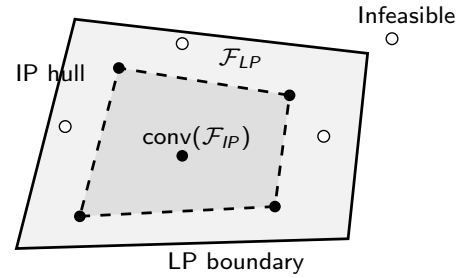


Figure: LP outcomes for $n = 8$ vertices using $\mathcal{LP}(G)$.



Set-theoretic



Geometric

The LP integral solution sequence $2, 3, 6, 10, 17, 29, 51, \ldots$ matches A285665. However, it is unclear why this should be the case.

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	3	0	1	0
4	5	1	5	0
5	7	4	20	3
6	13	15	107	21
7	22	55	767	200
8	44	263	9380	2659

Table: Outcomes of the LP relaxation for graphs of different orders n, classified according to whether a square root H exists.

The linear relaxation will never return an infeasible solution.

Proposition

If G has a square root H, then $\mathcal{LP}(G)$ is feasible.

Proof.

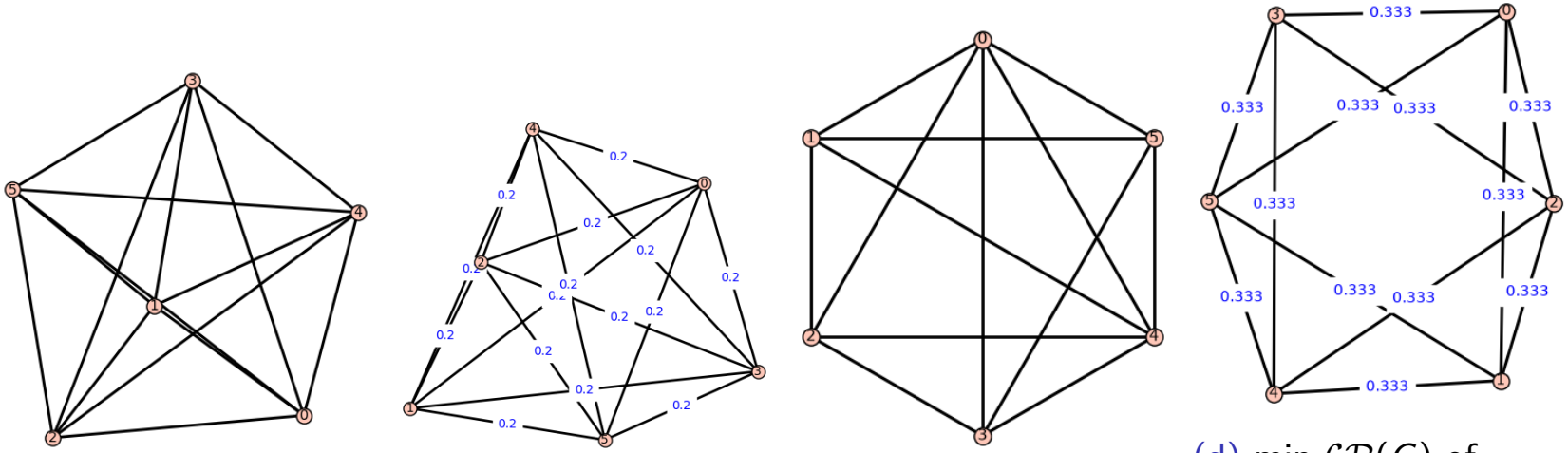
Since $\mathcal{LP}(G)$ is obtained by relaxing the integrality constraints of $\mathcal{IP}(G)$, every feasible solution of $\mathcal{IP}(G)$ is also feasible for $\mathcal{LP}(G)$. Hence $\mathcal{LP}(G)$ is feasible. □

Min and Max Objective Function

$$\min \sum_{i=1}^{n} \sum_{j=1}^{n} x_{ij}. \tag{23}$$

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	2	1	1	0
4	3	3	5	0
5	3	8	20	3
6	4	24	107	21
7	5	72	767	200
8	7	300	9380	2659

Table: LP outcomes for different values of n using the linear program with a minimisation objective function.



(a) K_6. (b) $\min \mathcal{LP}(G)$ of K_6. (c) $K_6 \backslash \{13, 25\}$. (d) $\min \mathcal{LP}(G)$ of $K_6 \backslash \{13, 25\}$.

Figure: For K_6, the solution to $\min \mathcal{LP}(G)$ is completely fractional, with every edge having weight $\frac{1}{5}$. For $K_6 \backslash \{13, 25\}$, the solution is completely fractional, with every edge having weight $\frac{1}{3}$.

Conjecture (Fractional Cliques in $\min \mathcal{LP}(G)$)

Let G be a graph with $n \geq 3$ vertices. For every edge $uv \in E_G$, let t_{uv} be the number of vertices in a smallest maximal clique of G containing uv. Then $\min \mathcal{LP}(G)$ *returns the solution given by*

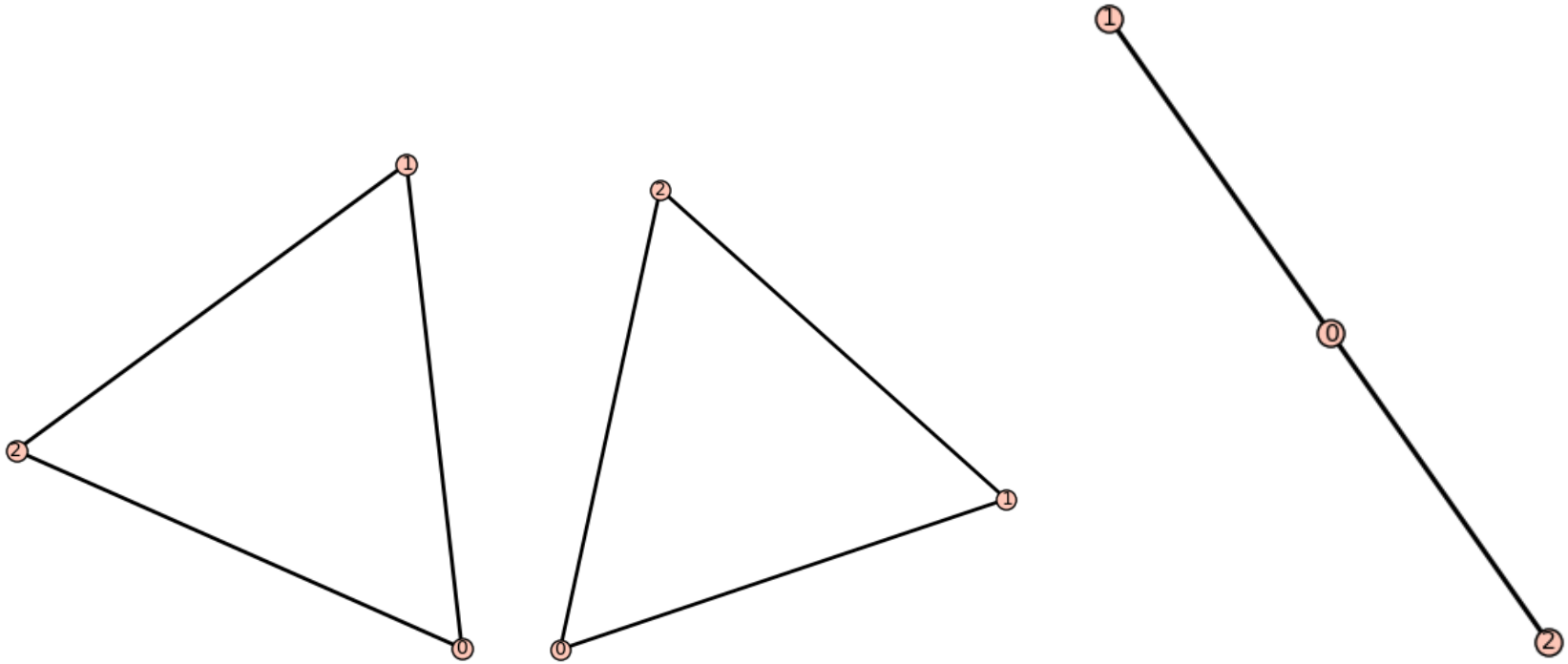
$$x_{uv} = \frac{1}{t_{uv} - 1} \quad \text{for all } uv \in E_G.$$

$$\max \sum_{i=1}^{n} \sum_{j=1}^{n} x_{ij}. \tag{24}$$

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	3	0	1	0
4	6	0	5	0
5	9	2	20	3
6	18	10	107	21
7	38	39	767	200
8	94	213	9380	2659

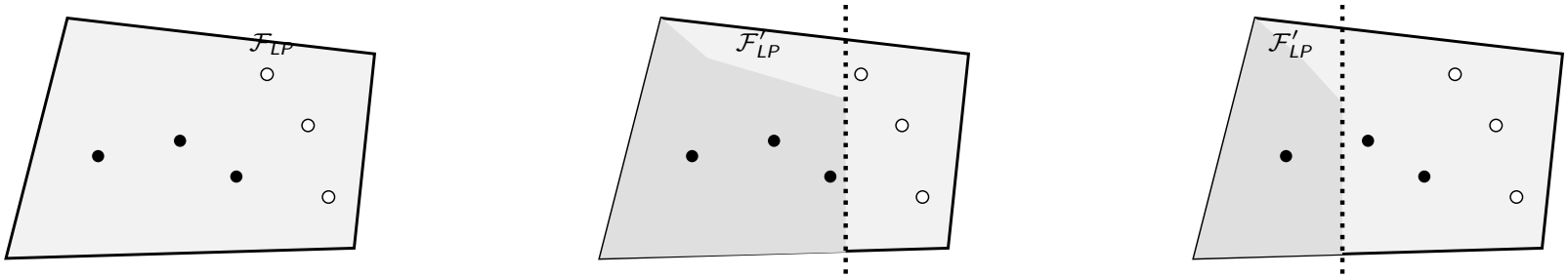
Table: LP outcomes for different values of n using the linear program with a maximisation objective function.

The single complete graph K_3 is selected as the unique optimal solution.



(a) K_3. (b) K_3 root of K_3. (c) A path root of K_3.

Figure: The graph K_3 and two of its square roots.



Base LP **Useful strengthening** **Incorrect strengthening**

A suitable constraint eliminates fractional solutions while retaining all valid integer solutions, whereas an overly restrictive constraint also excludes valid integer solutions.

Fundamental Properties

Lemma

Let G and H be graphs such that $G = H^2$. Let $a, b \in V_G$ such that $ab \notin E_G$. For every feasible solution x of $\mathcal{LP}(G)$ and every $v \in V_G$ with $v \neq a, b$ we have $x_{ab} = y_{avb} = 0$.

Lemma

Let G and H be graphs such that $G = H^2$. Let $a, b \in V_G$ such that $ab \notin E_G$. For every feasible solution x of $\mathcal{LP}(G)$ and every $v \in V_G$ with $v \neq a, b$ we have

$$x_{av} + x_{vb} \leq 1.$$

Lemma

Let G and H be graphs such that $G = H^2$. Let $a, b, v \in V_G$ such that $ab \notin E_G$. If a feasible solution x of $\mathcal{LP}(G)$ satisfies $x_{av} = 1$, then $x_{vb} = 0$.

Identifying Square Characteristics of Tree Roots

Definition (Sparse Pendant)

Let H be a graph. A **sparse pendant** is a path $s - u - v$ in H such that s is a leaf and u has degree 2.

Definition (Pendant Fork)

Let H be a graph. A **pendant fork** is the induced subgraph of H on four vertices s_1, s_2, u, v, where s_1 and s_2 are leaves adjacent to u, and every neighbour of u other than v is a leaf.

Lemma

Every tree with at least 3 *vertices contains either a sparse pendant or a pendant fork subgraph.*

Leaf Fixing Linear Program

If s is a leaf, u is its corresponding stem, and there exists $v \in N_G[u] \setminus \{u\}$ such that

$$N_G[s] \subset N_G[u] \subseteq N_G[v],$$

with all vertices in $N_G[s] \setminus \{u, v\}$ having the same closed neighbourhood as s, then the corresponding root edges can be fixed by imposing

$$x_{su} = x_{us} = x_{uv} = x_{vu} = 1. \tag{25}$$

Sparse pendant: s belongs to a unique K_3 in G.

Pendant fork: An induced $K_4 = \{s_1, s_2, u, v\}$, where s_1, s_2 are leaves of u, satisfies

$$N_G(s_1) \setminus K_4 = N_G(s_2) \setminus K_4 \subset N_G(u) \setminus K_4 \subseteq N_G(v) \setminus K_4.$$

Lemma

Let G and T be graphs such that $G = T^2$ and T is a tree. $\max \mathcal{UVLP}(G)$ always fixes a sparse pendant or pendant fork subgraph of T.

Whenever a root H of G exists, the addition of the constraint Equation (25) preserves the feasibility of the linear program.

Proposition (Validity of Unicliqual Vertex Fix)

If a root H of G exists, then $\max \mathcal{UVLP}(G)$ *is feasible.*

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	3	0	1	0
4	6	0	5	0
5	10	1	21	2
6	22	6	117	11
7	51	26	852	115
8	148	159	10271	1768

Table: Outcomes of the unicliqual vertex linear program for graphs of different orders n.

Propagation of the Fixing

Lemma (0 Path Propagation)

Let G and H be graphs such that $G = H^2$. Let $a, i, j, b \in V_G$ such that a, i, j, b form a shortest induced path in H between a and b. For every feasible solution x of $\mathcal{LP}(G)$,

$$x_{ai} = 1 \implies x_{ib} = 0, \quad x_{ia} = 1 \implies x_{bi} = 0, \quad x_{jb} = 1 \implies x_{aj} = 0, \quad x_{bj} = 1$$

Lemma (1-Path Propagation)

Let G be a graph and let x be a feasible solution of $\mathcal{LP}(G)$. Suppose that $u - v - a_1$ form a shortest induced path in H between u and a and that $x_{uv} = 1$. Let $A = \{a_1, \ldots, a_r\}$ and $B = \{b_1, \ldots, b_s\}$ be such that every vertex $k \in V_G \setminus \{u, v\}$ satisfies $k \in A \cup B$, and suppose that

$$x_{um} = 0 \text{ for every } m \in A, \quad g_{un} = 0 \text{ for every } n \in B,$$

and $g_{sa_1} = 0$ for the vertex s preceding u on the shortest path to a_1. Then

$$x_{va_1} = 1.$$

Vertex Deletion

Input: A graph G
Output: A LP model (m, x)
Construct and solve $\mathcal{LP}(G)$.
if $\mathcal{LP}(G)$ is integral **then**
    return (m, x)
end if
Set $F = \left\{ \{a, b\} \in \binom{V_G}{2} : x_{ab} = x_{ba} = 1 \right\}$.
Compute $d_F(u) = |\{w \in N_G(u) : \{u, w\} \notin F\}|$ for each $u \in V_G$.
Choose $v \in \arg\min_{u \in V_G} d_F(u)$ and set $V^- = V_G \setminus \{v\}$.
Construct $G^- = G[V^-]$ and inherit the fixing $F^- = \left\{ \{a, b\} \in F : a, b \in V^- \right\}$.
Construct and solve $\mathcal{LP}(G^-) + F^-$.
if $\mathcal{LP}(G^-) + F^-$ is integral **then**
    Construct H^- from its integral solution.
else
    return (m, x)
end if
Construct a new copy of $\mathcal{LP}(G)$ and impose $x_{ab} = x_{ba} = 1$ for every $\{a, b\} \in F$ and every $ab \in E_{H^-}$.
Solve the modified LP.
if the modified LP is infeasible **then**
    return (m, x)
else
    return the modified LP solution (m, x)
end if

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	3	0	1	0
4	6	0	5	0
5	11	0	21	2
6	25	3	117	11
7	62	15	852	115
8	183	124	10271	1768

Table: Outcomes of the vertex deletion linear program using unicliqual vertex as the base for graphs of different orders of n.

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	3	0	1	0
4	6	0	5	0
5	11	0	21	2
6	25	3	117	11
7	62	15	852	115
8	194	113	10271	1768

Table: Outcomes of the recursive vertex deletion linear program using unicliqual vertex as the base for graphs of different orders of n.

The linear program only finds a single solution, while there may be many distinct, non-isomorphic solutions. Hence, although fixing a subgraph obtained from the LP may be useful, the particular subgraph selected by the LP may not correspond to the subgraph required to continue the reconstruction of a square root.

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	3	0	1	0
4	6	0	5	0
5	11	0	21	2
6	26	2	117	11
7	73	4	852	115
8	274	33	10271	1768

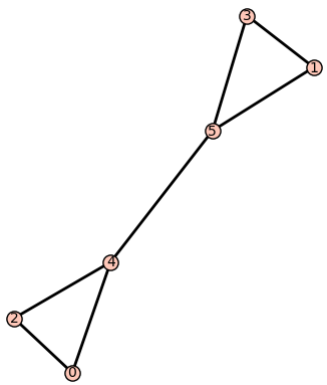
Table: Outcomes of the recursive vertex deletion linear program using unicliqual vertex as the base and enumerating all possible H^- roots for graphs of different orders of n.

Symbolic Sets

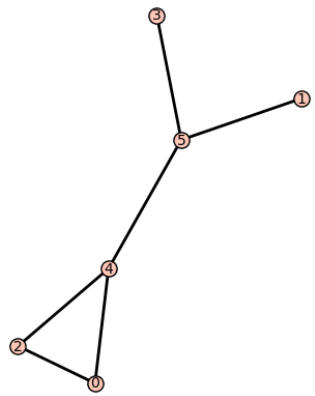
Conjecture

Let G and H be graphs such that $G = H^2$. Let $a, b \in V_G$ such that $ab \notin E_G$. For every feasible solution x of $\max \mathcal{LP}(G)$ and every $v \in V_G$ with $v \neq a, b$ we have

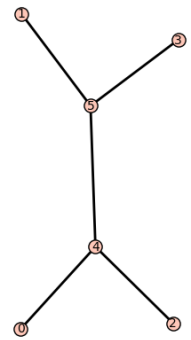
$$x_{av} + x_{vb} = 1 \text{ or } x_{av} + x_{vb} = 0.$$



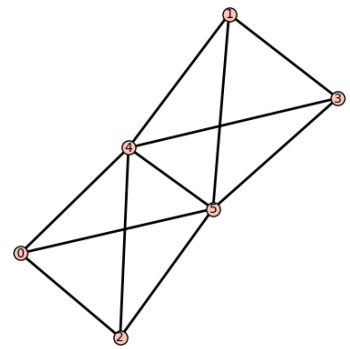
(a) The root graph G_1.



(b) The root graph G_2.



(c) The root graph G_3.



(d) The graph square $H = G_i^2$.

Figure: Three non-isomorphic square roots of the same six-vertex graph square.

Rounding